

Minimaps for Impossible Spaces: Improving Spatial Cognition in Self-Overlapping Virtual Rooms

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ABSTRACT

Natural walking in virtual reality is constrained by the physical boundaries defined by the size of the user's tracking space. Impossible spaces, a redirected walking technique, enlarge the virtual environment by creating overlapping architecture and letting multiple locations occupy the same physical space. Minimaps, which are small representations of the user's surroundings, are a common method to assist with wayfinding and navigation. Unfortunately, in a naive implementation of such minimaps for an environment with impossible spaces, the overlap would be obvious. We investigated approaches to displaying impossible spaces on minimaps without attracting users' attention to the overlapping parts of the VE. In this paper, we present a study that compared spatial cognition in a short gamified VR experience between three different conditions: (i) without minimap, (ii) with basic minimap, and (iii) with full minimap. Our results show significant improvements on accuracy in direction estimation tasks, suggesting that minimaps help users acquire spatial knowledge in the presence of impossible spaces.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Impossible spaces are a redirection technique that leverages self-overlapping architecture to build layouts that would be impossible in the real world [2] (see example in Figure 1). By making virtual rooms partially overlap with each other, the available virtual space can be enlarged without users noticing. However, if the overlap is too large, users will start to detect it [3].

A technique that has, to our knowledge, not yet been combined with impossible spaces are minimaps. Minimaps are a downsized two- or three-dimensional representation of the virtual world, akin to real-world maps.

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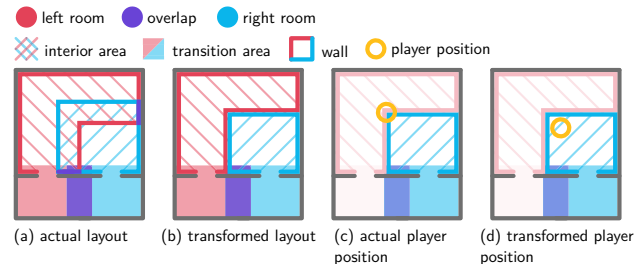


Figure 1: Transformation of overlapping rooms into a consistent representation for a minimap. In (c) and (d), only the blue room is visible and the user's position is translated accordingly on the minimap.

In commercial games, two-dimensional (2D) minimaps are common, and in VR, three-dimensional (3D) minimaps promise to improve user orientation and navigation skills [1].

Improving the acquisition of spatial knowledge, i.e. the *spatial cognition*, is desirable for VEs containing impossible spaces as well. However, if and how impossible spaces can be integrated with minimaps are still open questions. Simply showing overlapping geometry would probably reduce user's sense of presence, since it informs them about the impossible nature of the VE, which typically leads to breaks in presence.

We present a minimap that displays down-scaled versions of overlapping rooms, showing no overlap at all. At this point, it is an interesting question if the minimap still contributes to an improved spatial cognition or if the transformed minimap hinders acquisition of spatial knowledge. We report a preliminary user study examining the influence of our minimap on participants' spatial cognition. We compare participants' performance in estimating directions and search tasks under three conditions: (i) no minimap (No Map), (ii) a minimap without a location indicator (Basic Map), and (iii) a minimap with a location indicator (Full Map).

2 EXPERIMENT

The experiment was conducted as a remote study and requirements were a SteamVR-compatible HMD, an active internet connection and a tracking space of at least 2m by

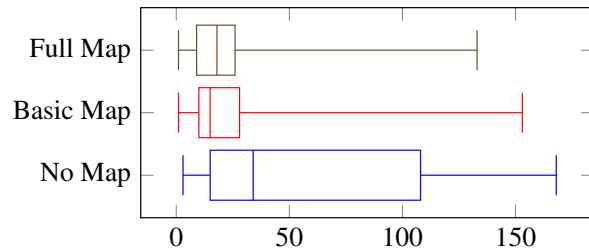


Figure 2: The quartiles of participants' pointing errors. The x-axis shows the angle of the error in degrees, the y-axis shows the condition.

2m. It was completed by 21 participants, who were students and members of the computer science department, or recruited via various websites¹. The total time per participant, including the introduction, tasks and survey, was 10-20 minutes. Participants wore their HMDs the entire time. They were not informed of the purpose of the experiment beforehand. We used a within-subjects design with participants completing two naive searches, two direction estimation tasks and one primed search for each of the three conditions.

The virtual environment consisted of 6 different floors using impossible spaces that were connected by a virtual elevator. The minimap was scaled to 5% of the size of the virtual world. The transformation process that turns overlapping rooms into a minimap representation is shown in Figure 1.

The 6 floors were split into three groups of two floors, and each group was randomly assigned one of the three minimap conditions. This assignment was evenly distributed among all participants.

The distribution of pointing errors with respect to targets' real positions is visualized in Figure 2. For each of the conditions, a Shapiro-Wilk test revealed that pointing errors with respect to targets' real positions were not distributed normally. Using a Friedman test at the 5% significance level, we found a significant effect of the minimap condition on the pointing error for real positions ($\chi^2 = 6.333, p = .042$). After checking for a symmetric distribution of differences, we ran a post-hoc Wilcoxon signed-rank test with Bonferroni correction. It revealed significant differences between no map and basic map conditions ($r = .421, p = .002$) as well as no map and full map conditions ($r = .421, p = .002$), but no significant differences between basic map and full map conditions for the pointing error on real positions.

The distribution of search times is shown in Figure 3. We tested the times users needed for the naive search tasks for normality using a Shapiro-Wilk test, which indicated

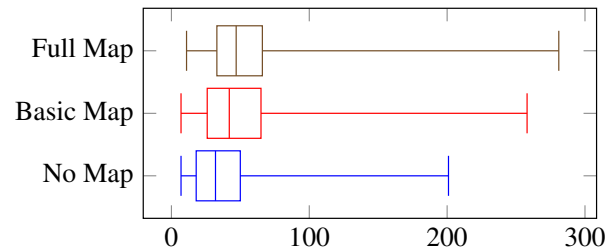


Figure 3: The quartiles of times participants needed to complete naive search tasks. The x-axis shows the time needed in seconds, the y-axis shows the condition.

that none of the results for all three conditions followed a normal distribution. Using a Friedman test at the 5% significance level, we found a significant effect of the minimap condition on the search time users needed for naive searches ($\chi^2 = 8.442, p = .014$). After checking for a symmetric distribution of differences, we ran a post-hoc Wilcoxon signed-rank test with Bonferroni correction.

It revealed significant differences between no map and full map conditions ($r = .391, p = .01$) but no significant differences between no map and basic map, or basic map and full map conditions for naive search time.

3 DISCUSSION & CONCLUSION

We found a significant improvement in pointing accuracy during a direction estimation task, suggesting that participants acquired more accurate spatial knowledge both with- and without a location indicator. Hence, it can be concluded that impossible spaces and a transformed minimap do not have to be an obstacle for spatial cognition. However, we also observed that users performed significantly worse with the full map condition, and slightly worse under the basic map condition. Although the latter is not significant, both results support the theory that the minimap did have a negative effect on naive search performance. The observed effect might be due to the additional time users needed to get an overview of the minimap and reconcile it with their mental model of the level.

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¹<https://www.xrdrn.org> and <https://www.reddit.com>